

CH365 Chemical Engineering Thermodynamics

Lesson 16 Sensible Heat Effects

Ethylene Glycol Process

Slide 2

- Sensible heat effects are associated with temperature change
- Latent heat (phase changes) – no temperature change
- Heat of Reaction
- Heat of Mixing

“Sensible” Heat Effects

Sensible = No phase transitions, no chemical reactions, and no change in composition.

$$U = U(T, V)$$

Total differential introduced in [L10](#), Slide 8, page 138

$$dU = \left(\frac{\partial U}{\partial T} \right)_V dT + \left(\frac{\partial U}{\partial V} \right)_T dV$$

0 for constant volume process, ideal gases, or incompressible fluids

[L11](#), Slide 4

$$dU = C_V dT$$

Eq. 4.1

$$Q = \Delta U = \int_{T_1}^{T_2} C_V dT$$

$Q = \Delta U$ for mechanically reversible, constant volume process (why? $\Delta U = Q + W$)

$$\frac{C_V^{\text{ig}}}{R} = \frac{C_P^{\text{ig}}}{R} - 1$$

Eq. 4.6

dimensionless

$$H = H(T, P)$$

0 for constant pressure process, ideal gases or approximately for real gases at high T and low P

$$dH = \left(\frac{\partial H}{\partial T} \right)_P dT + \left(\frac{\partial H}{\partial P} \right)_T dP$$

Need $C_P(T)$

$$dH = C_P dT$$

Eq. 4.2

$$Q = \Delta H = \int_{T_1}^{T_2} C_P dT$$

Eq. 4.3

steady-flow heat transfer

$$C_{P,\text{mixture}}^{\text{ig}} = y_A C_{P_A}^{\text{ig}} + y_B C_{P_B}^{\text{ig}} + \dots$$

Eq. 4.7

$$\frac{C_P^{\text{ig}}}{R} = A + BT + CT^2 + DT^{-2}$$

(constitutive – CH485)

Eq. 4.5

(dimensionless)

$Q = \Delta H$ for mechanically reversible, constant pressure closed-system processes or when ΔE_K , ΔE_P , and $W_s = 0$ in H-form of 1st Law eq 2.31

$$\Delta H + \frac{\Delta(u^2)}{2} + g\Delta z = Q + W_{\text{shaft}}$$

[L11](#), Slide 4

Mean Heat Capacity

Integral evaluated forms – “user-defined functions” – simplifies working with mixtures

These forms are used in later derivations in the textbook.

ICPH:
$$\int_{T_0}^T \frac{C_P}{R} dT = A(T - T_0) + \frac{B}{2}(T^2 - T_0^2) + \frac{C}{3}(T^3 - T_0^3) + D\left(\frac{T - T_0}{T \cdot T_0}\right) \quad \text{Eq. 4.8}$$

$$\Delta H = \text{ICPH} \cdot R$$

$$\int_{T_0}^T \frac{C_P}{R} dT = \left[A + \frac{B}{2}(T + T_0) + \frac{C}{3}(T^2 + T_0^2 + T \cdot T_0) + \frac{D}{T \cdot T_0} \right] (T - T_0)$$

MCPH:
$$\frac{\langle C_P \rangle_H}{R} = \left[A + \frac{B}{2}(T + T_0) + \frac{C}{3}(T^2 + T_0^2 + T \cdot T_0) + \frac{D}{T \cdot T_0} \right] \quad \text{Eq. 4.9}$$

$\langle C_P \rangle_H$ is the **mean heat capacity**.

$$\Delta H = \langle C_P \rangle_H (T - T_0)$$

$$\Delta H = \text{MCPH} \cdot R \cdot (T - T_0) \quad \text{Eq. 4.9}$$

From the mean value theorem of integral calculus:

$$f_{\text{avg}} = \frac{1}{b-a} \cdot \int_a^b f(x) dx = \langle f \rangle$$

Ideal Gas Heat Capacity in Simulators

Improved function – used in professional simulators such as CC and Aspen+

F. A. Aly and L. L. Lee, “Self-Consistent Equations for calculating the Ideal Gas Heat Capacity, Enthalpy, and Entropy,” *Fluid Phase Equilibria*, 1981, Vol. 6, Issues 3-4, pp. 169-179.

a, b, c, d & e are constants published and maintained by DIPPr (link on course web site).

$$C_p = C_p(T) = a + b \cdot \left(\frac{c/T}{\text{Sinh}[c/T]} \right)^2 + d \cdot \left(\frac{e/T}{\text{Cosh}[e/T]} \right)^2$$

hyperbolic sine hyperbolic cosine

DIPPr Eq 107

The image shows a sequence of three software windows from Aspen+ illustrating how to find the DIPPr equation 107 for ideal gas heat capacity:

- Component List**: A window showing a list of components. "1 (62) Water" is selected. A blue arrow points to the "View/Edit" button at the bottom.
- View/Edit Component Data**: A window with a sidebar on the left. A blue arrow points to "Heat Capacity Data" in the sidebar.
- Library Heat Capacity Data**: A window for "Water" (Component ID: 62). It displays three sections:
 - Ideal Gas Heat Capacity (J/kmol-K)**: The "Equation No." field is circled in red and contains the value "107". Below it are fields for "Low T (K)" (100), "High T (K)" (1500), "Low value" (33359), and "High value" (47105.1).
 - Liquid Heat Capacity (J/kmol-K)**: The "Equation No." field contains "100". Below it are fields for "Low T (K)" (273.15), "High T (K)" (533.15), "Low value" (76150.6), and "High value" (89393.9).
 - Solid Heat Capacity (J/kmol-K)**: The "Equation No." field contains "100". Below it are fields for "Low T (K)" (3.15), "High T (K)" (273.15), "Low value" (180.148), and "High value" (38120.5).
 To the right of these sections are coefficient tables. A large red circle highlights the coefficients for the Ideal Gas section:

Component ID:	62
Coefficients: A	33359
B	26798
C	2609.3
D	8888
E	1167.6
F	
G	

At the bottom of the third window, a blue arrow points to the "Help" button. Below the windows, the text "CC/DIPPr eq 107 found here" is displayed.

Ideal Gas Heat Capacity in Simulators

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$$C_p = C_p(T) = a + b \cdot \left(\frac{c/T}{\text{Sinh}[c/T]} \right)^2 + d \cdot \left(\frac{e/T}{\text{Cosh}[e/T]} \right)^2 \quad \text{DIPPr Eq 107}$$

hyperbolic sine
hyperbolic cosine

The screenshot shows the Aspen+ software interface. The 'Results - Pure Components' window is open, displaying a table of heat capacity data for Methane, Ethane, and Propane. The table has columns for Component, Temperature units, Source, and Property units. The data is organized into rows for Element 1 through Element 7. A red circle highlights the 'Element 1' row for Methane, which contains the value 33298. A blue arrow points to the 'T-Dependent' tab, and another blue arrow points to the 'CPIGDP-1' parameter.

Component	METHANE	ETHANE	PROPANE
Temperature units	K	K	K
Source	PURE37	PURE37	PURE37
Property units	J/KMOL-K	J/KMOL-K	J/KMOL-K
Element 1	33298	44256	59474
Element 2	79933	84737	126610
Element 3	2086.9	872.24	844.31
Element 4	41602	67130	86165
Element 5	991.96	2430.4	2482.7
Element 6	50	298.15	298.15
Element 7	1500	1500	1500

Aspen+/DIPPr eq 107 found here

Questions?